Aleph for TAK — Plugin Plan

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Summary

Build one server-side capture service from Aleph's code, and make the three plugins thin clients that draw an area, request a capture, and import the result as a TAK data package.

Aleph (MIT, Python 3.11, about 6,000 lines) lets you draw a rectangle or polygon and downloads everything inside it: satellite imagery, elevation, OSM buildings and roads, street-level photos, and a 3D mesh. It also answers single-point lookups from a CLI. In TAK, that becomes "draw a search area, get an offline data package of it," which fits pre-mission planning and field teams working without connectivity.

Three decisions shape everything below:

- **Server does the work, plugins stay thin.** Tiling, merging, GeoTIFF and DTED writing belong in Python on a server, not reimplemented three times in Java, C#, and TypeScript.
- **The data package is the contract.** Every client consumes the same zip, so ATAK and WinTAK get offline use for free through their existing import paths.
- **Aleph's Google sources must be replaced before any CAP use.** See the next section; this is the main blocker, not the plugin code.

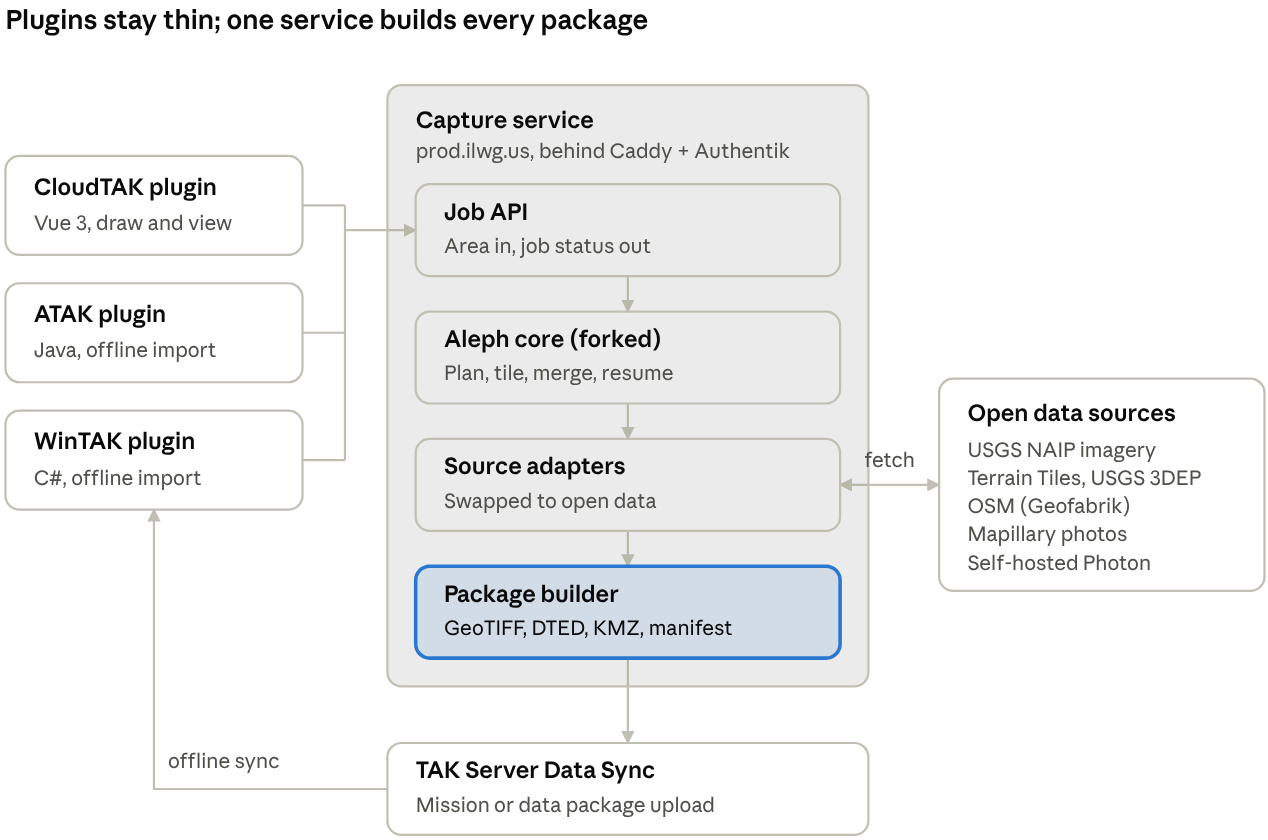
Blocker: replace the Google sources

Three of Aleph's five layers scrape undocumented Google endpoints, which the README itself warns may break or rate-limit. Bulk downloading and offline caching of Google imagery breaks Google's terms, and a CAP-branded tool should not depend on that. Keep Aleph's planning, tiling and merge code; swap the source modules.

Layer	Aleph source today	Problem	Replacement for TAK use
Satellite imagery	Google Maps tiles (`mt1.google.com`)	Undocumented; terms bar bulk and offline use	USGS NAIP (public domain, about 0.6 m, contiguous US); keep a pluggable WMTS/XYZ source for imagery CAP is licensed to use
Street-level photos	Google Street View internal APIs	Undocumented; terms bar caching	Mapillary API (CC BY-SA, needs a token) or Panoramax; mark coverage as sparse
3D mesh	Google Earth, reverse-engineered	Undocumented; no license	Drop from v1; revisit if a licensed 3D source appears
Elevation	AWS Terrain Tiles (Tilezen)	Open, attribution required	Keep; add USGS 3DEP 1 m where available; output DTED for ATAK
Buildings, roads	OSM via Geofabrik extracts	Open (ODbL), attribution required	Keep; output KML or KMZ with attribution
Place lookup	Photon (komoot public instance)	Public instance is fair-use only	Self-host Photon or Nominatim on the capture server

Aleph is MIT, so forking is fine; keep its copyright notice and the attribution files for each source.

Architecture



capture service architecture · 3 clients, 1 service

Each plugin sends an area and a layer list to the Job API. The forked Aleph core plans and downloads through open-data adapters, the package builder writes one zip, and TAK Server Data Sync pushes it to ATAK and WinTAK for offline use. CloudTAK reads results straight from the service.

The service can also run on a field node with a pre-cached area, so captures work when the main server is unreachable.

Platform plans

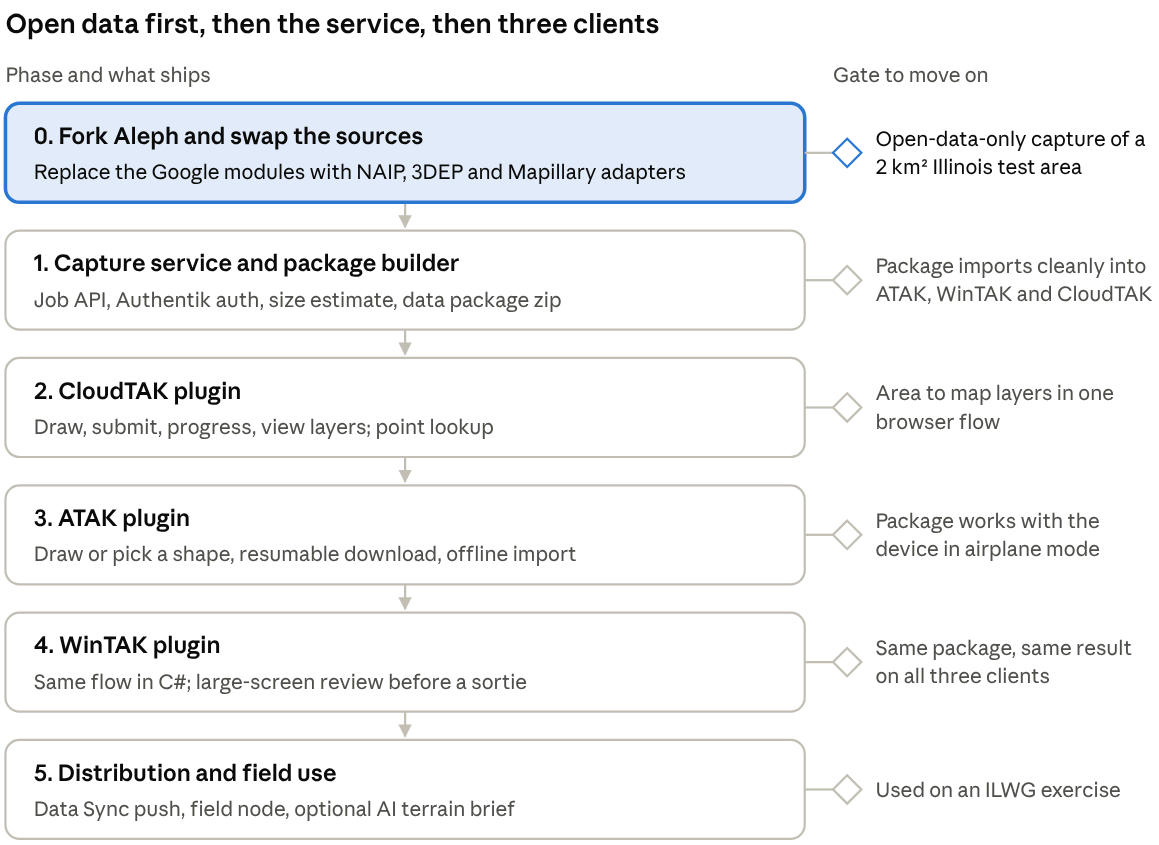
All three plugins share one flow: draw or pick an area, choose layers, see a size estimate, submit, watch progress, import the package. Point mode (tap a spot for elevation, nearest address, nearby photos) is the second feature on each.

	CloudTAK	ATAK	WinTAK
Stack	Vue 3 and TypeScript plugin	Java plugin on the ATAK-CIV SDK	C# plugin on the WinTAK SDK (.NET Framework)
Area input	Map draw tool, or an existing feature (LPB search ring, ICS 204 segment)	Existing drawing tools: rectangle, polygon, or a selected shape	Existing drawing tools or a selected shape
Import path	Load layers into the web map; offer the zip for download	Built-in data package import; DTED to the elevation store	Built-in data package import
Offline use	None; browser needs the server	Full, after the package syncs	Full, after the package syncs
Unique work	Reuses your CloudTAK plugin patterns; quickest to prove the service	Download over spotty LTE: resumable, Wi-Fi-only option, storage check	Large-screen review of the package before a sortie
Distribution	Bundle with the ILWG CloudTAK deploy	Signed APK through TAK.gov third-party pipeline for Play Store ATAK	Installer through TAK.gov or direct to ILWG machines

Package contents for v1: imagery as a cloud-optimized GeoTIFF (plus MBTiles if ATAK draws it faster), elevation as DTED and GeoTIFF, OSM features as KMZ, photos as point placemarks with the image attached, and a manifest. Test each format on the ATAK and WinTAK versions ILWG runs before locking it in.

Build order

Phase 0 is the critical path: no plugin is worth shipping until the service captures from legal sources.



build order · 6 phases, each with a gate

CloudTAK goes first because it needs no SDK signing and reuses your existing plugin patterns, so it proves the service fastest. ATAK comes before WinTAK because field teams carry phones, not laptops.

Risks and open questions

The biggest risk is imagery quality: NAIP is free and legal but refreshed every two to three years, so a licensed source may be needed for recent damage or construction.

- **Package size.** A few square kilometers of 0.6 m imagery plus DTED is tens to hundreds of MB. Show the estimate before submit and cap area per request.
- **Upstream breakage.** Aleph's author ignores backward compatibility by design. Fork once and own the code; don't track upstream.
- **Street-level coverage.** Mapillary is thin in rural Illinois, where most searches happen. Treat photos as a bonus layer.
- **SDK churn.** ATAK and WinTAK plugins are tied to app versions. Pin the versions ILWG fields and rebuild on each upgrade.

- **Data handling.** Captures of search areas can reveal mission details. Keep packages behind Authentik and expire them after the mission.

Decisions needed:

- [ ] Ship as its own plugin, or as a module inside FeatureLink?
- [ ] Which imagery is CAP licensed to use beyond NAIP?
- [ ] Host only on prod.ilwg.us, or also on a field node for offline captures?
- [ ] Add an AI-written terrain brief of each package (Aleph's original purpose), with human review, or leave it out?

Sources

- Belluxx/Aleph on GitHub: README, `docs/cli.md`, and source modules, read 2026-10-22
- Data-source endpoints and licenses were taken from Aleph's own source and attribution notes; TAK SDK and format support should be confirmed on ILWG's fielded app versions